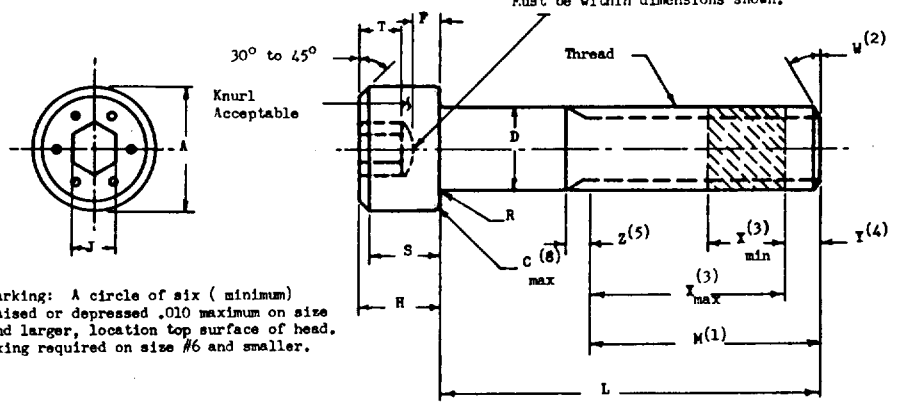
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Head Marking: A circle of six ( minimum) dots raised or depressed .010 maximum on size 8-32 and larger, location top surface of head. No marking required on size #6 and smaller.

(D) REINSTATED AFTER 16 AUGUST 1982.

| THREAD (6) | D Body Dia. MAX - MIN | A Head Dia. MAX - MIN | M Thread Length (1) MIN | X (3) MIN | (7) ULTIMATE TENSILE STRENGTH (Wtu) LBS. MIN. | H Head Height MAX - MIN | J Socket Width Across Flats MAX-MIN | T Key Depth MIN | R Radius MAX-MIN | F Hole Depth Limit MIN | S Head Side Height MIN |
|---|---|---|---|---|---|---|---|---|---|---|---|
| #4-40 UNC-3A | .1120 .1075 | .183 .176 | .750 | .125 | 960 | .112 .108 | .0952 .0937 | .051 | .009 .005 | .009 | .101 |
| #6-32 UNC-3A | .1380 .1329 | .226 .218 | .750 | .156 | 1,450 | .138 .134 | .1111 .1094 | .064 | .010 .006 | .013 | .124 |
| #8-32 UNC-3A | .1640 .1585 | .270 .262 | .875 | .156 | 2,280 | .164 .159 | .1426 .1406 | .077 | .012 .007 | .011 | .148 |
| #10-32 UNF-3A | .1900 .1840 | .312 .303 | .875 | .156 | 3,200 | .190 .185 | .1587 .1562 | .090 | .014 .009 | .016 | .171 |
| 1/4-28 UNF-3A | .2500 .2435 | .375 .365 | 1.000 | .178 | 5,820 | .250 .244 | .1900 .1875 | .120 | .014 .009 | .026 | .225 |
| 5/16-24 UNF-3A | .3125 .3053 | .468 .457 | 1.125 | .208 | 9,280 | .312 .306 | .2530 .2500 | .151 | .017 .012 | .027 | .281 |
| 3/8-24 UNF-3A | .3750 .3678 | .562 .550 | 1.250 | .208 | 14,000 | .375 .368 | .3160 .3125 | .182 | .020 .015 | .029 | .337 |
| 7/16-20 UNF-3A | .4375 .4294 | .656 .642 | 1.375 | .250 | 18,960 | .437 .430 | .3790 .3750 | .213 | .023 .018 | .031 | .394 |
| 1/2-20 UNF-3A | .5000 .4919 | .750 .735 | 1.500 | .250 | 25,600 | .500 .492 | .3790 .3750 | .245 | .026 .020 | .051 | .450 |
| 5/8-18 UNF-3A | .6250 .6163 | .937 .921 | 1.750 | .278 | 41,000 | .625 .616 | .5050 .5000 | .307 | .032 .024 | .055 | .562 |

(1) Thread length. The length of the screw thread is measured from the extreme point to the last pitch of complete thread. On short screws the complete (full formed) threads shall extend to within two thread pitch lengths of the head.

(2) The point shall be flat and chamfered. The flat shall be normal to the axis of the screw and the chamfer (W) shall be at an angle of 35° 40' - 0° with the plane of the flat. The chamfer shall extend slightly below the root of the thread, and the edge between flat and chamfer shall be slightly rounded.

(3) X min represents the minimum length of external thread required for engagement with complete internal thread pitches. The element shall engage within this minimum length and meet requirements of MIL-F-18240. X min is equal to 5 thread pitches. X maximum equals the length of complete thread between Y and Z.

(4) Y min shall have at least one pitch of complete thread. Y maximum equals 2 pitches of complete thread plus 2 pitches which includes the incomplete thread and chamfer. The locking element shall not be effective within the area of Y minimum.

(5) Z equals incomplete thread and/or extrusion angle permissible up to 2 thread pitches length maximum. The element or any machine holes or grooves for the element shall not penetrate this area.

(6) Threads shall be fully formed by any single rolling process, and in accordance with specification MIL-S-7742.

(7) Ultimate Tensile Strength ($W_{tu}$) is based on 160,000 psi and stress area ($A_s$) = $3.1416\left(\frac{E}{2} - \frac{3H}{16}\right)^2$.

| | | |
|---|---|---|
| P.A. Navy - AS<br>Other Cust<br>Air Force - 11 | TITLE<br>SCREWS, SELF LOCKING, 250°F, CYLINDRICAL HEAD HEXAGONAL WRENCHING SOCKET, ALLOY STEEL, 160 KSI Ftu (Socket Head Cap Screws) | MILITARY STANDARD<br>MS21262 |
| PROCUREMENT SPECIFICATION<br>FF-S-86 ' MIL-F-18240 | SUPERSEDES<br>MS21262(ASG) | SHEET 1 OF 2 |

DD FORM 1 SEP 87 672-1 (Limited coordination)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

APPROVED 9 June 1960 REVISED (D) 7 April 1961 (B) 20 Feb 63 (C) 15 September 1981 (D) 16 August 1982

NOTICE.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

This standard has been approved by the Buldep, Department of the Navy, and the Department of the USAF and is mandatory for use by that activity. All other military activities are required to employ this standard where suitable.

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(8) Radius or Chamfer (45°) Depth is .005 for #4 thru #10; .008 for 1/4 thru 3/8; and .010 for 7/16 thru 5/8.

MATERIAL: Alloy Steel; Prpcurement Specification FF-S-86.
HARDNESS: RC36 - 45 up to and including 1/2 inch, RC36 - 43 over 1/2 inch.
PLATING: Cadmium Plating, QQ-P-416, Type II, Class 3.
SURFACE ROUGHNESS: The surface of the shank, fillet and bearing area shall not exceed 125 in accordance with MIL-STD-10.
LOCKING ELEMENT: In accordance with Specification MIL-F-18240. Locking element area must pass thru a plain ring gage, equal to the maximum major thread diameter + .010 inch, freely or with finger pressure.
DESIGN AND USAGE LIMITATIONS: See MS 15991.
DIMENSIONS: In inches.
EXAMPLE OF PART NUMBER: MS 21262-56 = 1/4-28 Screw, 1 inch long.

| (B) L SCREW LENGTH | DASH NUMBERS THREAD SIZE AND PITCH | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 4-40 | 6-32 | 8-32 | 10-32 | 1/4-28 | 5/16-24 | 3/8-24 | 7/16-20 | 1/2-20 | 5/8 -18 |
| .1875 ±.015 | (1) 1 | 2 | 3 | | | | | | | |
| .250 ±.015 | 4 | 5 | 6 | | | | | | | |
| .3125 ±.015 | 7 | 8 | 9 | 10 | 11 | 12 | | | | |
| .375 ±.015 | 13 | 14 | 15 | 16 | 17 | 18 | | | | |
| .4375 ±.015 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | | | (1) |
| .500 ±.015 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | | | |
| .625 ±.015 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | | | |
| .750 ±.015 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | | | |
| .875 ±.015 | | 47 | 48 | 49 | 50 | 51 | 52 | | | |
| 1.000 +.031 -.015 | | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 |
| 1.250 +.031 -.015 | | | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 |
| 1.500 +.031 -.015 | | | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 |
| 1.750 +.031 -.015 | | | | 78 | 79 | 80 | 81 | 82 | 83 | 84 |
| 2.000 ±.031 | | | | 101 | 85 | 86 | 87 | 88 | 89 | 90 |
| 2.250 ±.031 | | | | | 91 | 92 | 93 | | 94 | |
| 2.500 ±.031 | | | | | | 95 | 96 | | 97 | 98 |
| 3 ±.031 | | | | | | | 99 | | 100 | |

(1) MIL-F-18240 establishes performance requirements for the element within X min. Therefore, only positive indication of torque is required for screws having lengths above this heavy line.

**CANCELLED AFTER 15 SEPT 1981. NO SUPERSEDING STANDARD.**

This standard takes precedence over documents referenced herein.
Referenced documents shall be of the issue in effect on date of invitations for bid.

APPROVED 9 June 1960 REVISED (D) 7 April 1961 (B) FOR CHANGES SEE SHEETS 1 AND 2. (C) 15 September 1981 (D) 16 August 1982

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| P.A. Navy - AS<br>Other Cust<br>Air Force - 11 | TITLE<br>SCREWS, SELF LOCKING, 250°F, CYLINDRICAL HEAD HEXAGONAL WRENCHING SOCKET, ALLOY STEEL, 160 KSI Ftu (Socket Head Cap Screws) | MILITARY STANDARD<br>MS21262 |
|---|---|---|
| PROCUREMENT SPECIFICATION<br>FF-S-86 & MIL-F-18240 | SUPERSEDES<br>MS 21262(ASG) | SHEET 2 |

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